

PREEMAS MC

ORNAMENTAL HORTICULTURE AND LANDSCAPE DESIGN CPT 442

Ornamental plants are plants that are grown purposely for landscaping of an area; it can be shrubs, trees or flowers.

^{importance}
~~importance~~ of ornamental plants

- i. Ornamental plants are important for interior and exterior decoration
 - ii. They are used for decoration of roads
 - iii. Some beautiful flowers are used in certain ceremony, eg wedding and burial ceremony
 - iv. Ornamental plants are source of income to the florist
- Landscape design is defined as the use of plant outdoor to fulfill aesthetic and functional purposes. It creates a visual representation of a site using scale dimension

Aim of Land Scope Design

- i. To enhance the aesthetic appeal of an area
- ii. To enhance the neighborhood and increase property value
- iii. Produce privacy by shielding the general public from selected areas such as the back yard of houses serves as windbreak (modify environmental factor)
- iv. Create recreational ground to provide place for relationship and community interaction
- v. Provide hobby activities for home owners
- vi. Improve and conserve natural resource by reducing soil erosion
- vii. Reduce noise and environmental pollution
- viii. Provide ample shade and coolness for people

Categories of landscaping

- i. Residential landscape: This landscape is fear toward individual house owner and neighborhood needs.
- ii. Public landscaping: here plant are use to enhance the frontage of offices and to beautify the indoor of offices and open spaces, e.g of public place that need to be landscape include parks, plazas city square and other recreational factories.
- iii. Commercial landscaping: This is the type of landscaping found in parks, shopping, hotels, banks and restaurant such business display plant in their various form to avoid abstracting the view in those areas.
- iv. Specialty landscaping: This kind of landscape is found in places such as botanical garden and zoological gardens where final design are often used. A biological garden is design to established a large variety of plants types, their design usually has a strong educational component plant species often being intensified by their common and scientific names.

THE BASIC ELEMENT OF LANDSCAPE DESIGN

Beauty is influence by culture, tradition and personal experience, the element that creates *beauty* can be learned. The landscape designer uses plant as the primary object in creating a design, plant species have certain features that influence how they are used in a landscape. This five element must be considered in designing both the hardscape and the soft scape of your property, the element are as follow:

1. Colour, 2. Form, 3. Line, 4. Scale, 5. Texture

1. Colour: tends to truly capture the eye of viewer. It is a key element in landscape design. When incorporating colour, it is important to consider the season. While flower and plant in varying houses make a wonderful summer garden. You will

also want to include vegetation that would offer colour during the other season, such as ever green, conifers and holly.

People respond differently to colour, choice and arrangement in the landscape are critically classification of the overall virtual appeal.

2. Form:

Form refer to the shape of the plant or hard scape feature. Plants available in many forms, upright, ground cover, round, and free form. Hard scape form can be in various height and shape. Plant of different forms may be grouped and arrange in a certain fashion to create a different overall form in the landscape, e.g a landscape design may group a number of plant with narrow column form to create a ~~horizontal~~ ^{vertical} form. This strategy is used in creating a hedge around a house. Plant have form because they have visual mass and thus occupy visual space form is therefore directly related to one's ability to see it.

3. THE ELEMENT OF LINE

The effect of line is accomplished through the arrangement of object, line is a boundary element in design shape and structure are defined by line. The aesthetic of colour is line, value and intensity.

Arrangement of landscape object in a line can be use to divert viewers or to the feel point on the design, line do not have to be straight and when they are straight, they do not have to be continuous. A designer can deliberately introduce break to ~~Save the~~ ^{Save the} viewer to pause and change viewer line. Line can be of different direction to slow designer viewing and to encourage the viewers to spend more time for appreciating a particular display.

SCALE OR PROPORTION

Scale or proportion is not only designed with respect to the plants in landscape but also in terms of their relationship with other structure and, the general

functionalities of the designs. In choosing plant for landscape, it is important to take into consideration their size at maturity. Since Inanimate object such as statues, sculptures and houses maintain their size. it is important that trees located in ^{the} vicinity do not over shadow and make them inconspicuous when they are fully grown. *groom*

ELEMENT OF TEXTURE

Plant texture refers to an objects feel with respect to the sense of touch, visual surface characteristic may project a soothing image or a harsh, coarse feeling of the viewers. Texture refers to how the surface of the object is perceived, relative to the objects around it we often rely on the relative size of a plant's leaves to draw conclusion about it perceive.

Features of Categories of ground/gardens

Garden is specialized area, set aside for the growth and development of ornamental plants and vegetables

School Garden/Primary and Secondary

School ground or garden requires a space in which children can roam freely, trees often dominate the landscape with some functional plants such as shrubs and a limited area for annuals or bedding plant with species which should be resistant to mechanical treatment. The gold of landscaping school ground is to achieve a low maintenance and child proves area

2. College Campuses

Unlike primary and secondary, high school college campuses can often afford gardens maintenance services, the aim of landscaping college campus is to provide *viewed* certainty by linking the diverse structure on site by using trees, ground covers and shrubs. College campuses landscaping is such an important part of each

identity, it can help them to embrace their gold or initiative such sustainability. College ground can even influence students to ^{the college} exercise one school over the other.

Recommendation for Campuses College Garden

- i. A thick belt of large shady tree should be planted to bring down noise and cut down dust and storms.
- ii. Frontage should be planted with medium sized flowering tree for beautification

Lawn grass also look nice in an educational premises like campus

- iii. Tree should not completely abstract the view of the building from outside.

PARK AND RECREATIONAL AREA

The key goal in designing a landscape for parks and recreational areas for public use is for public safety, appropriate plants should be selected so that the park remains open without hiding place that may encourage criminal activities, such park may have some significant number of non-plant materials such as bushes, playground, status plant etc are also used to complement trees and are shrubs grouped and spaced randomly as effective way of landscape park.

Cemeteries

Cemeteries, burial ground, grave site or grave yard is a place ^{where} remains of dead people are buried or otherwise intend. The word cemetery implies that the land is specifically as a burial ground.

FEATURE OF A REST GARDEN

Cemetery are scars due to the special and scars nature that cemetery hold in the societies, they landscapes are often reserve out of respect for the deceased, with the presumption that cemetery will provide eternal resting place for the dead. Cemeteries are delineated by fences of hedges which is less likely to happen with

paths

Lawns. They require higher level of landscape maintenance than park, to ensure that the grave is well cared for

A well-kept lawn is the primary landscape element in the modern cemeteries with a few trees planted in an informed design. Trees should preferably have horizontal forms.

Principle of landscape design

1. Balance- is a state of being as well as seeing. We are most comfortable in the landscape that have sense of balance. Using this principle, landscape designer creates landscape that are pleasing to look at and even inviting.
2. Focalization- any good design has a focal point. The place where the viewer's eye is first attracted. Focalization is some refers to as simply focal point. The focal point is the strongest of the design in any given view. A home's focal point is often the front door. The landscape focal point is often close to the front to the entrance of the home.
3. Simplicity- as the name implies simply mean keeping landscape simple, a landscape that make people happy and comfortable by avoiding using too many colours, shape texture.
4. Proportion- refers to the size relationship of all the features in the landscape. This include vertical, horizontal and special relationship. Proportion in landscape design extend to building size, lot size, plant size, area of planting and area of open space.

7 RECESS AND STAGES OF LANDSCAPE DESIGN

1. Site inventory and analysis

Preparation of a base plan using information taking from the plot of survey and actual measurement. Inventory of the site for the type and condition of existing

Plant

elevation, plot material, soil and other site feature, such as topography and important views, location of public utility, boundary line and easment location'

1. Design concept

Foundation of design idea and creative of scale ground plant and sketch defining the overall land scape design for the property. Client perception and suggestion are always taken into consideration.

2. Design development.

Master planning revisions of drawing after approve of the concept plan and preparation of other necessary drawing such as layout of hard surface and structure determine dimension and material to be used. Detail plant list—specifying plant by botanical name, common name, and quality size based on the design, colour and photograph of the plant used.

3. Technical documentation.

Construction of drawing details, specification tender documents and pretender cost estimate

Project administration

Prior to construction review and report necessary to insure that the work is being carried in accordance with the design and contract documents.

Classification of ornamental plant based on foliage and flower

1. Foliage Ornamental Plant: These are ornamental plant that are use full

because of the showing coloration, shape texture or arrangement of the foliage [leaves collectively as produced or arrange by nature] they are use in landscape scheme for emphasis in border and to lead individuality to group where specimen plant are used. eg acalipha, croton, caladiums, amaranthus species etc.

2. Flower Ornamental: These come in wide variety of shape, size and colour
Ornamental flowering plant are the center piece of any garden or land scape.

[The central or the most important feature] eg of ornamental flower are monkey flower, petunias flower, sunflower, morning glory, marigold etc.

Classification of ornamental plant base on life cycle

1. **Annual Ornamental Plant:** These are ornamental plant that grows fast and bloom quickly so as to set seed and produce. Annual plant complete their entire life cycle in one year going from seed to plant to flower and to seed then dried off.

2. **PERENNIAL ORNAMENTAL PLANT**

~~These on the other hand~~ are plant that go from seed to seed within one season but do not die at the end of the season. By definition, is a plant that you can expect to live for at least three or more years and are usually perennial inhabit.

3. **BIANNUAL PLANT**

These are plant that are in between annual and perennial by categories, it is regarded as short lived perennial that usually take two growing season to complete their life cycle the categories of the plant produces only foliage in the first year and flower and set seed in the second year.

PROPAGATION OF ORNAMENTAL PLANT

Propagation is the process of creating new plant from a variety of species such as ~~seed~~ **Cutting**, sucker, leaves and other plant parts.

There are two method of propagation of ornamental plant.

1. Sexual propagation
2. Vegetative propagation

Sexual propagation: Whereby seed are use for propagation through sowing seed are typically produced from sexual means because genetic recombination has take place. While

Vegetative Propagation: is the process of creating new plant from a variety of ~~seeds~~ budding, grafting cutting, layering or any other technique for vegetative propagation.

Technique for Vegetative propagation are:-

- Layering - **Cutting**
- Grafting
- Suckers
- Striking or cutting
- Budding

Table 1. Some ornamental plant and their mode of propagation

S/N	Ornamental plant	Flower colour	Mode of Propagation
1.	Thuja (Morphanhi)	Chrimson red brown	Seed/air layering
2.	Acalipha Wilkesiana	Various colour	Cutting
3.	Acalipha hispida	Green	Cutting
4.	Thuja accidentalist	Yellow	Seed
5.	Ixora coccinea	Rose pink	Sucker

Table 2. Classification of ornamental plant base on trees, shrubs and flowers

S/N	Ornamental trees	Ornamental shrubs	Ornamental flower
1.	Neem cassia	Allamanda	Justica
2.	Royal palm	Crotons	Canalili
3.	Frangi pani	Hibiscus	Morning glory
4.	Balsam	Wild rose	Sun flower
5.	Indian almold	Cauli flower	Mary gold
6	Elame of the forest	Acalipha	Camellina

Nursery production

Nursery is place or establishment for raising and handling of young seedlings until they are ready for transplanting on the permanent field

Factors to be consider in nursery layout

Just as in locating a greenhouse, nursery site should be carefully selected,

considering both the economic and ecological factors. These factors include;

Climatic factors such as;

- a. **Temperature:** Nurseries produces young plant or seedling and thus are sensitive to changes in temperature. Seedling are tolerant to rapid change in temperature. If a nursery is being considered for a hot environment, there is need to construct a structure that will provide or reduce the effect of such high temperature on the young seedling and vice-versa.
- b. **Rainfall:** The rainfall pattern of an area should be well understood as certain operation cannot be delayed in a nursery due to lack of water rainfall can be supplemented with irrigation if need be, areas that are more to severe weather should be avoided as storms may damage the young seedling.
- c. **Light:** Unlike in greenhouse in which supplemental light can be provided, field production relies solely on sunlight for the plant physiological activities such as photosynthesis, plant in the nursery should be provided with or exposed to adequate sunlight for their proper growth.

EDAPHIC SOIL FACTOR

- a. **Air pollution:** nursery should be located in areas where plants will not suffer from air pollution if it has to be located in a heavily industrialized area, it should be located up wind to escape the pollutants from various industrial facilities.

- b. Topography: Site for nursery establishment should be flat at terrace has implementations on drainage and soil erosion, ease of land preparation and General ease production operation if site is rough or sloppy, use of machinery may be difficult and soil erosion will be more pronounced.
- c. Drainage: The propose site for a nursery shall be naturally well drained and if not artificially drainage should be provided adding to the production cost. Drainage is required for good aeration of the soil and to reduce the incidence of soil borne disease.

- d. Soil texture: Soil texture and structure, since seedling need to be dug up at some point either bare root or ball or earth, a soil with loose structure is preferred, the soil should be cohesive enough to form ball around the roots. A proper textured soil also drain well, hold moisture at a desirable amount and is easy to work and well aerated.

- e. Supplemented water supply: A nursery operation cannot be 100% rainfall, for timely production water must be provided when it is needed by the plant depending on the location.

Pruning

Pruning: Is the removal of any undesirable part of the plant either branches, shoot, leaf etc that can stop the plant from its normal growth.

Objectives of pruning

- i. To control diseases and pest
- ii. To regulate or control the size of the plant
- iii. To maintain good appearance and to keep plant healthy
- iv. To remove no productive part of the plant
- v. To allow light and air penetrate the interior part of the plant.

TRANSPANTING

As the transferred of seedling from the nursery to it permanent filed. In transplanted raise bed are prepared in area where rainfall is heavy while sunken bed or flat bed may be used in area where the rainfall is light.

Transplanting is mostly done in the morning or late evening and should be transplanted with ball of earth.

Mulching: Mulching is the use of dried leaves or grasses round the base of plant to conserve moisture.

OBJECTIVES

- i. To conserve moisture
- ii. To control soil erosion
- iii. To avoid decomposition of soil organism matter.
- iv. To encourage the activities of soil organisms
- v. To add nutrient to the soil.

STAKING:

Staking is a way of supporting plant with either a stick or wood so as to make the plant stand upright and to prevent the branching from touching the ground.

IMPORTANT OF STAKING

- i. Staking is very important in supporting the tender shoot to stand upright.
- ii. To uplifting branching fruit on the soil to avoid soil contamination and infection.
- iii. There is also control of climatic heating to better crop.

FLOWER ARRANGEMENT

Ornamental plant play a vital role in the beautification of an areas such as institution, hotels, hospital and other building flower arrangement therefore is an importance part of ambience on any occasion such as wedding, birthday or any other event in life. It is an organization of design and colour towards creating an ambience using flower foliage and other floral accessories, flower arrangement are not only reserved to be used for occasion but can be used on the kitchen table in their own way as decoration, flower arrangement beautify homes and offices and are used as center piece for a table.

CLASSES OF FLOWER ARRANGEMENT

- i. Floral Bouquet: Here flower are organized in attractive way, so that it carried to a ceremony and present. These are simple and easier to assemble than a formal flower arrangement they are made of roses and are most popular.
 - ii. Centerpiece: Centerpiece are flower design to be place on tables when dinner are hosted on such occasion like thanks giving day or Christmas.
 - iii. Floral basket: Floral basket are flower arrangement done in basket of varied depth, they are very popular as gift especially on occasions basket may be filled with various kinds of festival fruit and vegetables.
 - iv. Floral wreath: A floral arrangement in form of a circular garland, they are usually use in traditional occasion to indicate honour, wreath are very much used on occasions like thanks giving day and Christmas to welcome guest when dinner are hosted.
- Shrubs are generally considered as a multi-stem wood plant that is less than 15 feet tall. It is an evergreen ornamental plant mostly for beautification of an environment.

FUNCTION OF SHRUBS

- i. Shrub are often place at the base of a house to hide or to serve as boundary between partners or (as fencing material)
- ii. Shrubs are used for the construction of houses.
- iii. Every green shrubs are used throughout the year to impact a solid feeling to the environment.

SHELTER BELT

These are planted row of tree or shrubs design to prevent erosion from weather. Shelter belt are great way minimizing damage and keep your property safe.

IMPORTANT OF SHELTER BELT

- i. Shelter belt is very important because it built a strong barrier between wind and your properties.
- ii. Shelter belt can reduce or prevent wind erosion by intercepting wing consequently reducing wind speed.
- iii. Planting trees for wind break is not only practical but is a way to add some beauty to your property.

ROOF GARDENING:

Roof gardening is known as terrace gardening the existing of roof (top can be effectively utilized for growing fruit plants' vegetable, piece, flower plant and ornamental plant, the explosion occurs every day resulting in migration of people from rural area to urban area for income Generation due to the migration of people, most of the agricultural land are converted into residential areas resulted with decrease population of fruit and vegetable.